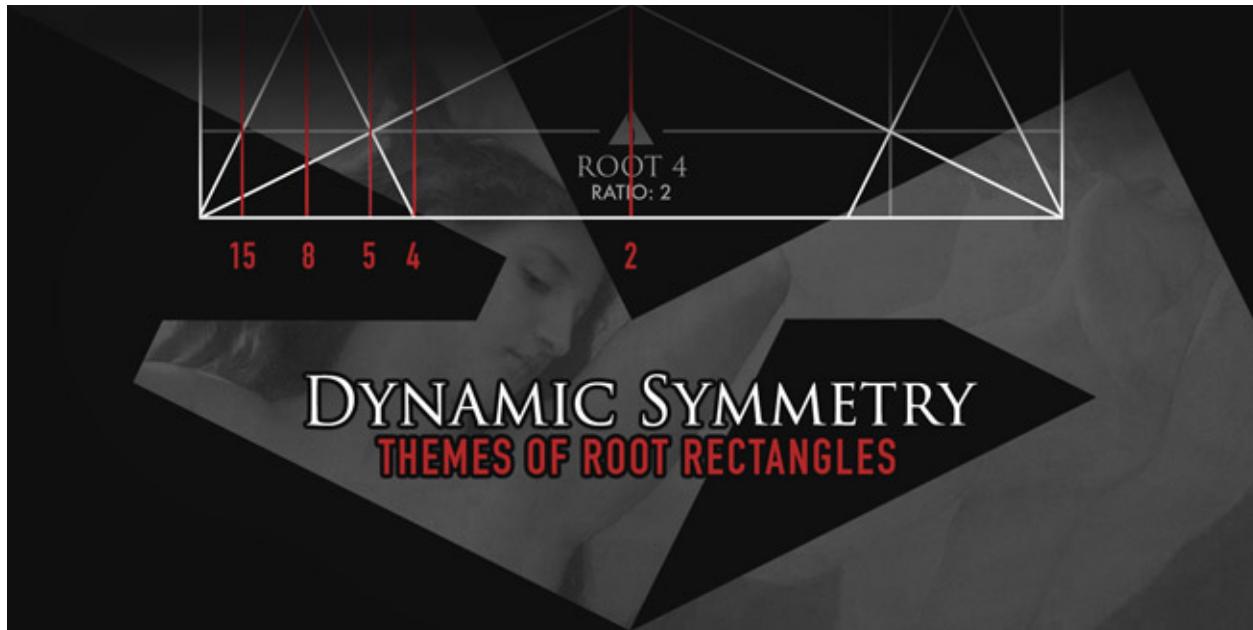




Dynamic Symmetry: Themes of Root Rectangles

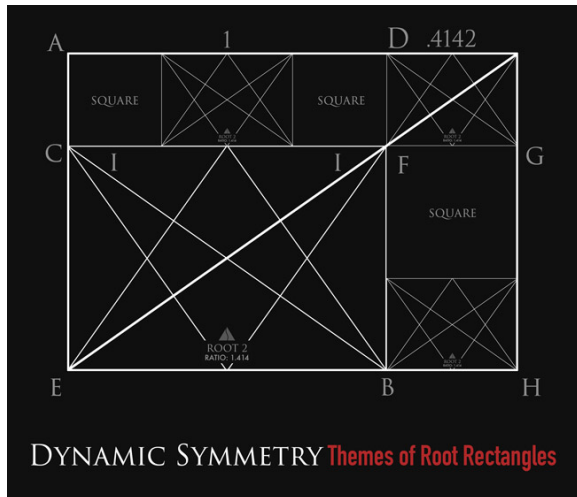
Welcome back everyone, thanks for all of the continued support! I've got a huge article for you today. We are going to be digging deep (super deep) into **dynamic symmetry** and looking at how the root rectangles can be easily divided into **themes**.

Dynamic symmetry is a simple tool...after all it's just lines right? Yet, a lot of us are confused by it...I know I was when I first started learning about it. There's not a specific point you need to start (unlike the rule of thirds). It's not to be thought of as restricting, but to **inspire** the organization of a **masterful composition** with unity, rhythm, and movement. The piano is limited to 88 keys, but do composers like Mozart or Ludovico Einaudi complain? No, they use this limitation as **power**, just like a writer uses grammar and sentence structure.

This article is **advanced knowledge** of dynamic symmetry (see [Day 14](#) for prerequisite), so if you've been wanting to learn more...here ya go! No geometric equations or crazy math skills required, we are just drawing lines.

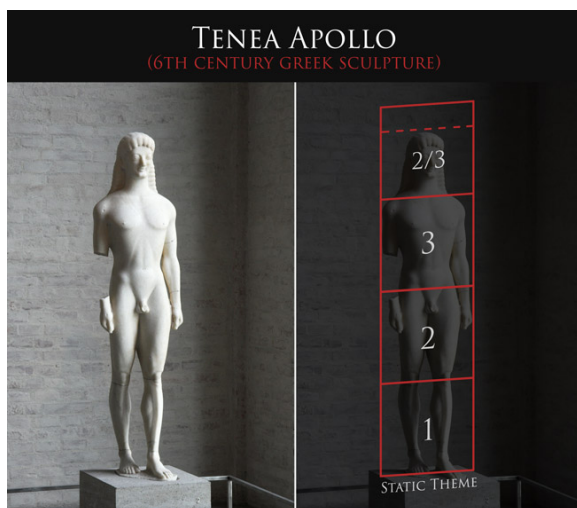
Defining the Term: Theme

The term "theme" is used by Jay Hambidge to describe the **area divisions within a rectangle**. This could be countless different arrangements. On page 45 of the book "[The Elements of Dynamic Symmetry](#)," we see an example like the illustration below. Jay Hambidge describes it as, "Every area in the root-two rectangle AH is now exhausted and the entire rectangle divided into a **theme** of squares and root-two rectangles."



Hambidge also explains themes to be in a **static** class, or static theme. In his book he describes the following image as, "The [Tenea Apollo](#), a 6th century B. C. Greek statue, has a symmetry **theme** of three and two-thirds. This means that if a rectangle is made, the area of which consists of three squares and two-thirds of a square, it will be exactly the shape which will contain the projection of the statue. When we find that every member of the body is expressible in terms of this shape and that the theme produces a simple pattern form throughout we decide that this Apollo belongs to the static class."

The statue below is proof that the Greeks didn't have knowledge of the Egyptians **secret** symmetry at the time it was made. They learned of dynamic symmetry around 6th century B.C, after the statue was made, and it quickly replaced the static symmetry they were using. It's amazing to read that this dynamic symmetry stuff was being used by the Egyptians around the 1st Dynasty. Wow!



Numeric Dynamic Themes (NDT)

As with all of the **terminology** for dynamic symmetry (baroque diagonal, sinister diagonal, reciprocal, etc), it helps us describe the tool we are using in our art to easily communicate it in a learning environment. Jay Hambidge uses the term themes in his book, which we covered

already. Myron Barnstone mentions (in his [Golden Section DVD](#) series) the themes of two and three when referring to the root two rectangle; neither of them have covered it in the way I demonstrate throughout this article. So, get excited to learn something new!

Since there aren't any terms that I know of for what we are learning today, I'll define it as "**numeric dynamic themes**" or **NDT** for short. I name it this because we aren't creating complex themes as demonstrated earlier by Jay Hambidge, but rather, themes easily defined by the **vertical divisions** of the **basic armature** (see [Day 14](#)).

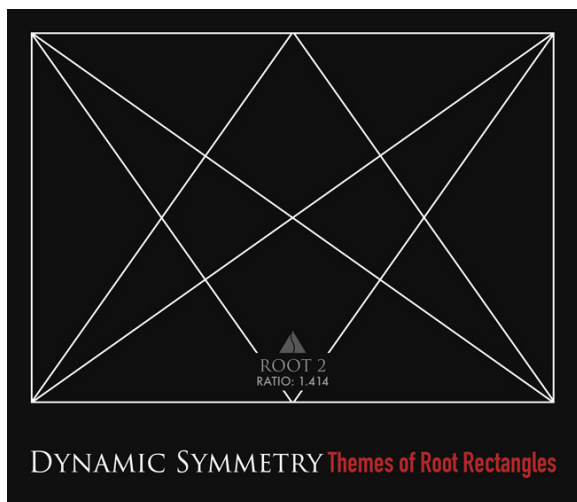
The area divisions of the numeric dynamic themes are just smaller root rectangles inside the mother. The thing that makes them special is that they are equally aligned to the vertical divisions. If you're lost, just hang in there because we are going to see plenty of examples.

Who Would Use This Stuff?

Learning the NDT of each root rectangle is great if you're a painter, illustrator, graphic designer, pattern maker, sculptor, or digital artist. If you are a photographer you will most likely never use it past the theme of two (aka **Major Area Divisions**). If you're a photographer doing some composite work of multiple photos, then this information will be beneficial to learn. The themes are used to refine art to the smallest detail...not something typically used for photography....but we should never rule it out.

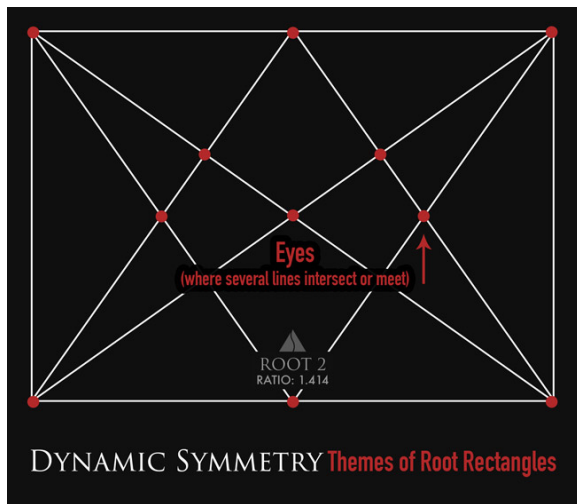
Root Two Rectangle - Ratio: 1.414

We'll learn the basics of numeric dynamic themes by studying the root two rectangle first. The grid below should look very familiar to you. It's the basic armature of the root two rectangle...minus the horizontals and verticals.



Finding the Eyes

More basics for you...the intersection or meeting point of multiple lines is called an "eye." In the diagram below you can see eyes being created on the edges and within the rectangle. You can use these points to generate more lines to be used for your composition. Master painters like **Lautrec** (see [Day 79](#)), **Picasso** (see [Day 103](#)), and **Degas** (see [Day 194](#))...all used the eyes to generate more lines to compose their masterpiece.

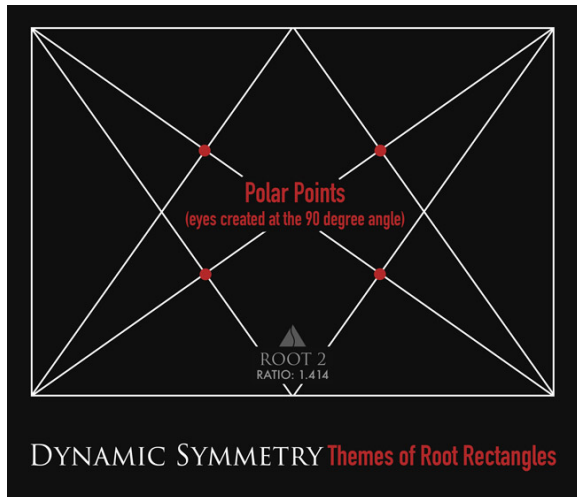


Here are the **analyzed paintings** showing how the masters used the eyes to generate more lines for their composition.



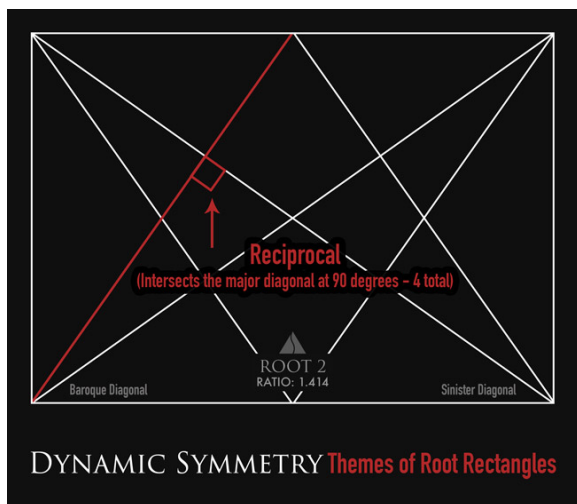
Finding the Polar Points

The **polar points** (see [Day 5](#)) are the eyes created by the 90 degree angle...where the reciprocal meets the major diagonal. These points are **special** in many ways when constructing our grid. Not in a way that people *believe* the rule of thirds is special (please...don't get me started haha!). The polar points are special from a geometric stand point. They are the eyes which generate the horizontals and verticals we use in our basic armature, and they are the points which spiral equal proportions of rectangles around each other. The rectangle of the whirling squares (phi) is a great example of this.



Finding the Reciprocal

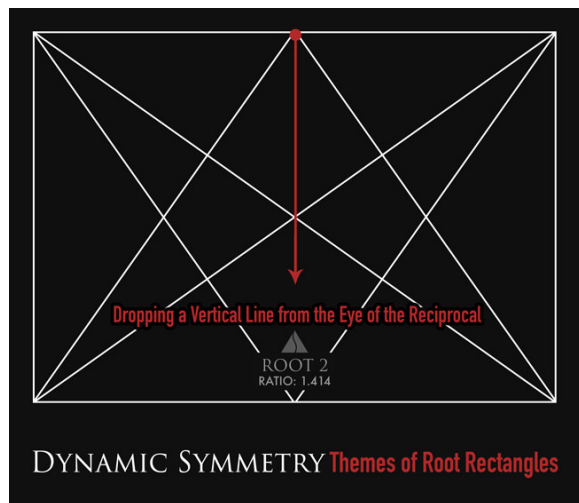
This is probably more review for you veterans, but it's great to demonstrate for the juicy steps to follow. The reciprocal is created by running a line from the corner and intersecting a major diagonal at 90 degrees. There are two major diagonals (baroque and sinister) and four reciprocals in every rectangle.



Dropping a Vertical Line to Complete the Reciprocal

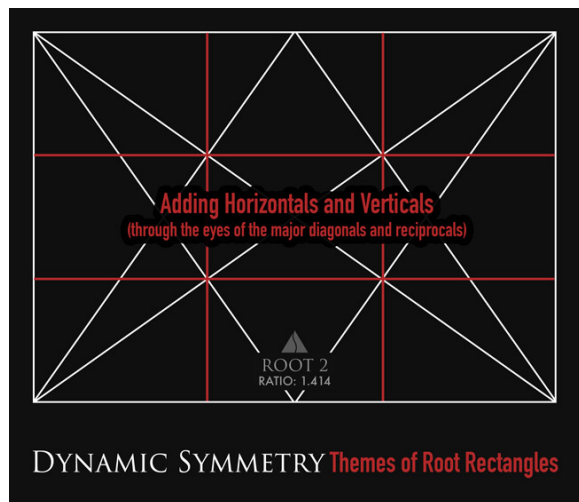
To complete the reciprocal all we have to do is drop a vertical line down from the eye that meets the edge. This is a key move to generating one of the themes. We'll learn more as we progress through the diagrams.

Keep in mind that the root two is the only one that is divided in half when the reciprocal line is dropped....pretty handy when wanting to divide things up evenly.



Adding Horizontal and Verticals

As mentioned before, the polar points are considered to be special in many ways...geometrically speaking. These simple horizontals and verticals created from the polar points will generate the eyes for our NDT! Exciting right? And easy!



Vertical Lines Creating Maximum Themes

If we use these eyes generated by the horizontals, verticals, and diagonals of our basic armature we get the locations for our NDT. We can start by dropping a line down in the center and working left or right. We only need one half, so in this case I generated the lines on the left...that's why you don't see any red lines on the right side of the example below.

We can see that we now have a theme of two, three, four, and six. This means for the theme of two we can fit two smaller root two's inside the mother and they will meet at the line labeled "2." We can fit three root two's inside the mother and they will meet at the "3," four at "4," and six at label "6."

Sure, we could squeeze **five** or **twenty** smaller root two rectangles inside the mother, but it wouldn't line up *exactly* to these vertical lines generated by the eyes of our basic armature.

We can **shorten** the writing of each rectangle and theme used with the following characters:

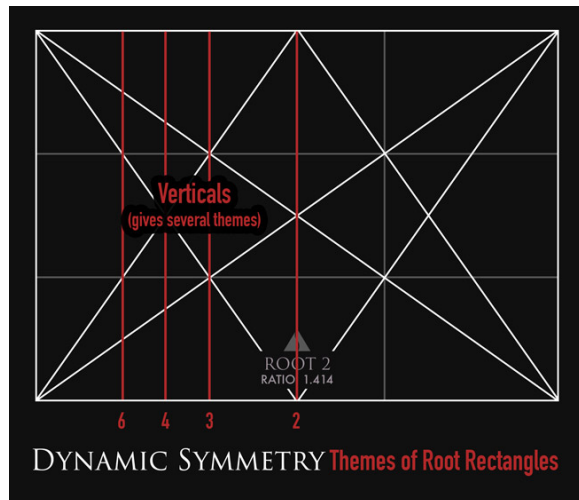
Root 2 on the theme of two = $\sqrt{2}:2$ or r2t2 (for file naming)

Root 2 on the theme of three = $\sqrt{2}:3$ or r2t3

Root 2 on the theme of four = $\sqrt{2}:4$ or r2t4

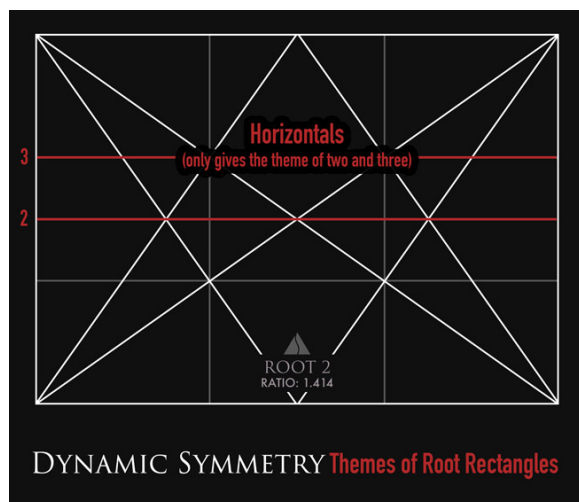
Root 2 on the theme of six = $\sqrt{2}:6$ or r2t6

Note: to get the root symbol on a Mac press "Option"+ "V"



Horizontals Creating Limited Themes

To test all of the areas, I ran horizontal lines through the eyes of the basic armature and got fewer themes. In the case of the root two rectangle we only get the themes of two and three. So, in all root rectangle we will have the **theme of two**, and the **plus one** of the root rectangle. What is the plus one? This means the root three will have a theme of two and a theme of four (the plus one) when running the horizontals or verticals. The root five rectangle will have a theme of two and a theme of six when running the horizontals or verticals.

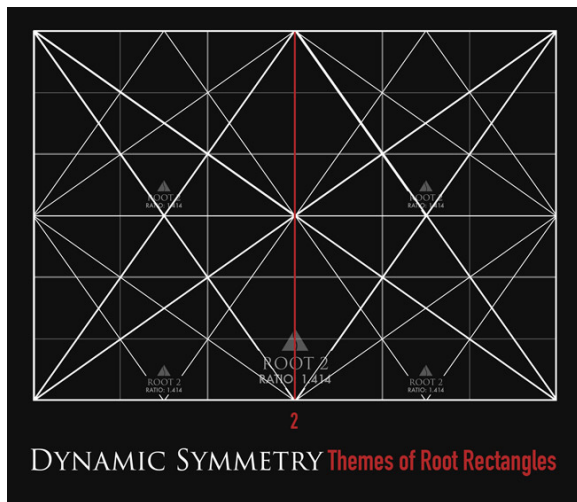


Root Two Rectangle: Theme of Two

$\sqrt{2}:2$

Now I'll start showing you what the numeric dynamic themes look like when constructed. Every rectangle, arbitrary, phi, or root, will have *at least* a theme of two. The diagonals of any rectangle will always create a center point for our theme of two.

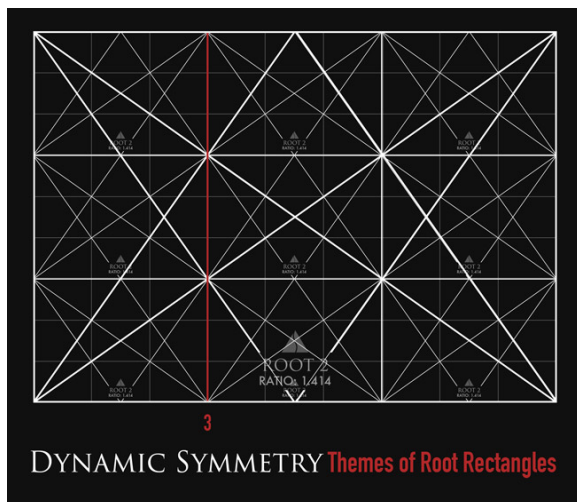
We can see that it is a theme of two, but there are actually four smaller root two's inside the mother. Each theme will follow...the same amount of rectangles vertically and horizontally. If it's on the theme of twelve it will have twelve smaller root rectangles vertically and twelve smaller root rectangles horizontally...totaling 144 (12 x 12) smaller root rectangles within the mother.



Root Two Rectangle: Theme of Three

$\sqrt{2}:3$

Here we see the root two on the theme of three, totaling 9 smaller root two rectangles within the mother. It's going to start getting easier to understand from here on out because the other root rectangles are constructed in the same manner.

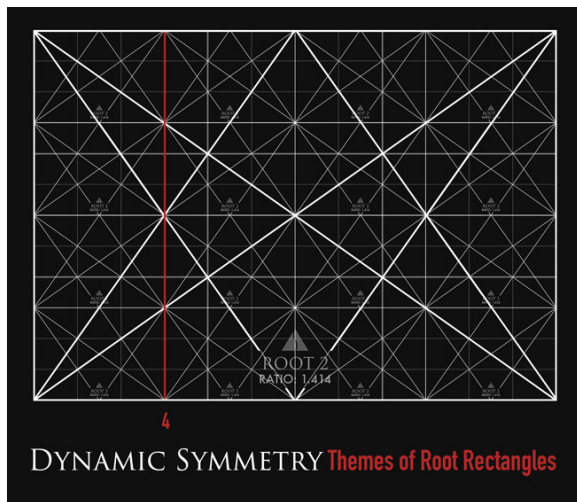


Root Two Rectangle: Theme of Four

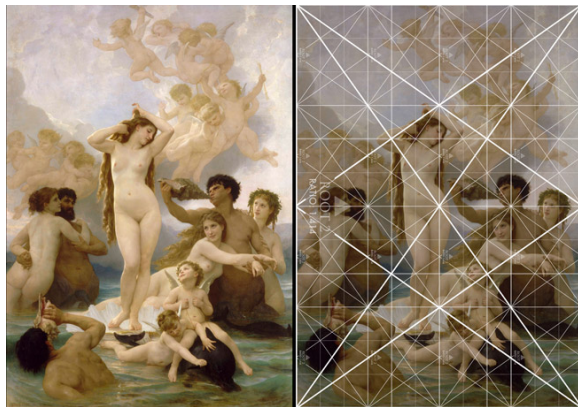
$\sqrt{2}:4$

You may be noticing that these multiplied themes are similar to the previous ones. The theme of four is similar to the theme of two.

If it has a theme of two, then it will have a theme of four. We are talking about all of the *root rectangles*, but the phi, root phi, and 1.5 are exceptions to this.



Here's an amazing painting by Bouguereau (see [Day 65](#)) that shows how he uses the root two on the theme of four. How do I know which theme he used? I lined up all of the themes (2, 3, 4, 6) and the theme of four **locked in** the elements more exact than any other. Of course, we could get similar results if we used the theme of three and continued to **parallel** the diagonals and generate lines with the eyes. Using the theme of four seems more efficient for his finished design.

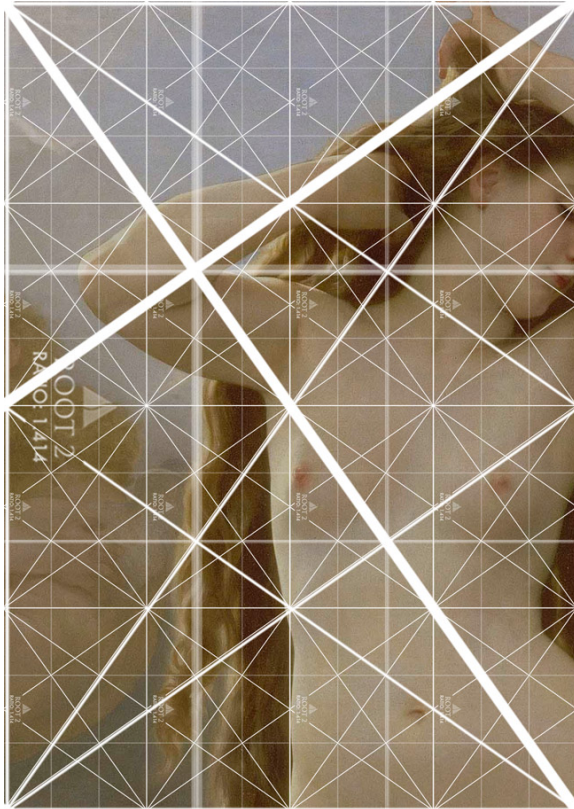


Once you get familiar with dynamic symmetry and building the themes within the armature, you can break it down even further to help compose your design. We can see in these next detail images of Bouguereau's masterpiece that he is using a smaller root 2 on the theme of four to find the details of his figure. The entire canvas doesn't have to be filled with grids, he just builds the armature where he needs.

We can see how he's locking in the hand, chin, neck, eye, nipple, shoulder, belly button, torso, elbow, and hair.

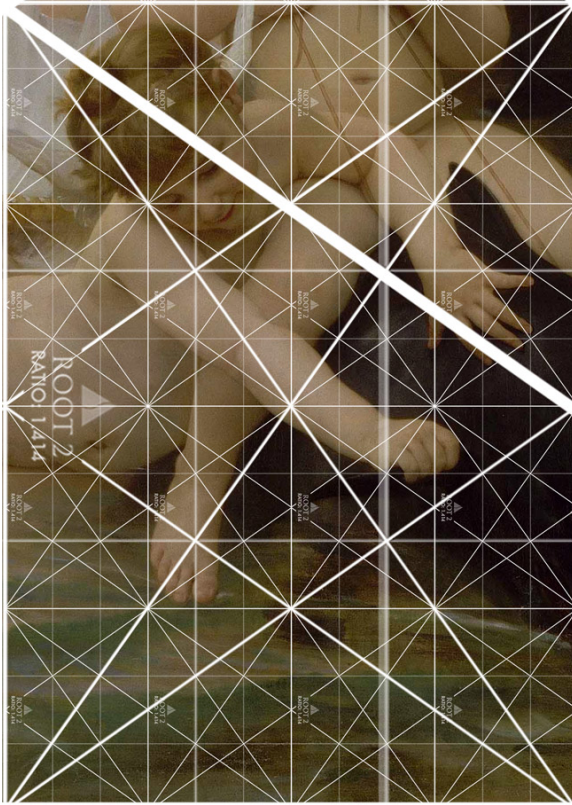
Can We Combine Themes?

Now, you may be wondering if we can use a root two on the theme of four, but use a smaller root two on the theme of three for the smaller details like this. Certainly! We shall bind ourselves to no rules! If it helps your design then go for it, but for now I would recommend keeping it in the same theme like Bouguereau...why make things more difficult? Keep it in the same theme, then generate more lines through the eyes if necessary.



In this detail, we see him locking in the arms, hand, eyes, foot, and head. Sure, you could **argue** that the more lines we put on there, the more things will look like they are lining up. This argument comes up all the time, but to **debunk** this, all we have to do is look at how the **diagonals** line up...exactly. Look at the arm...it's exactly the same diagonal. This isn't arbitrarily lining up...it was planned and calculated...designed with mastery.

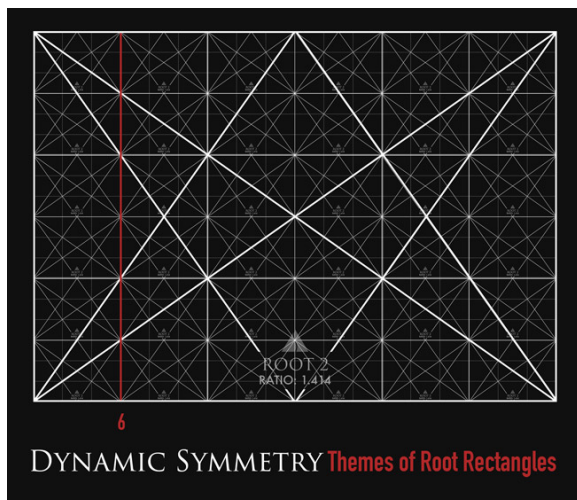
This is found throughout every master painting I've analyzed. So **eye-opening** and amazing!



Root Two Rectangle: Theme of Six

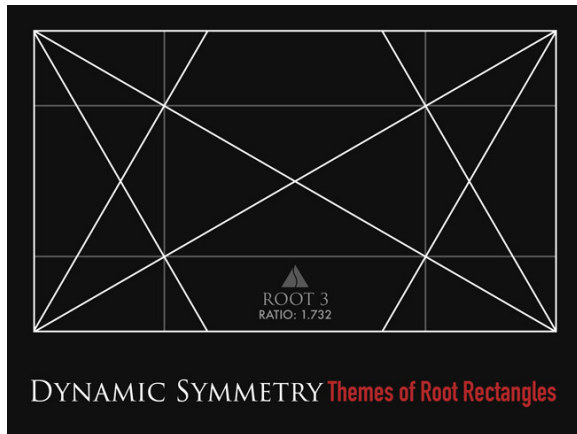
$\sqrt{2}:6$

Are you starting to see how the vertical lines generated from the eyes of the basic armature are creating the NDT?



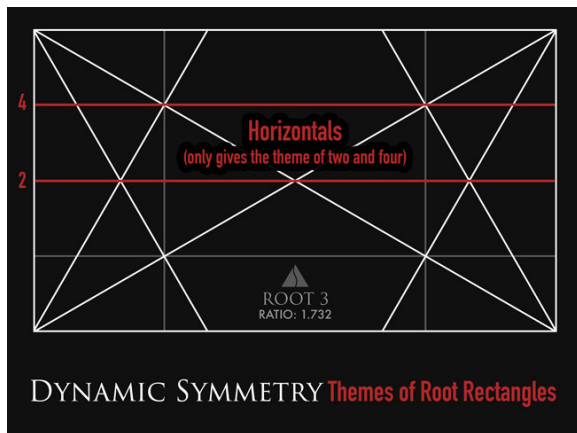
Root Three Rectangle - Ratio: 1.732

Now, let's run through the rest of the root rectangles and see how we create the numeric dynamic themes.



Root Three Horizontal Themes

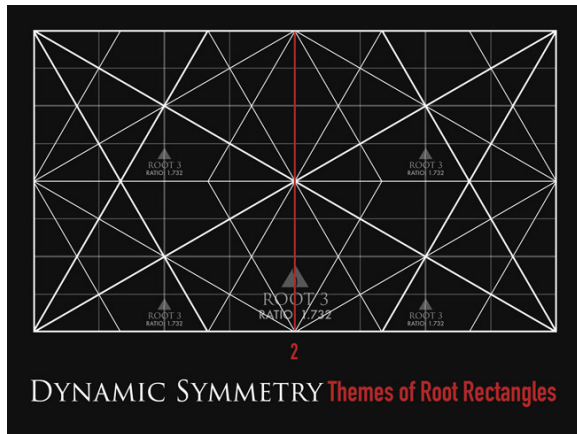
I'll test all of the root rectangles with the horizontals just to further understand how they create fewer themes than the verticals.



Root Three Rectangle: Theme of Two

$\sqrt{3}:2$

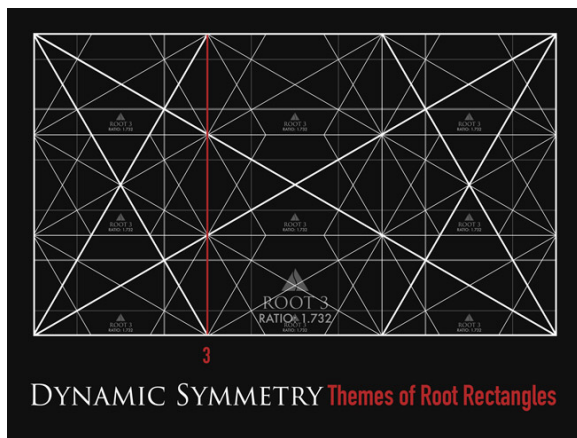
As mentioned before, all of the rectangles will have a theme of two, and all of the root rectangles will have a theme of two and four.



Root Three Rectangle: Theme of Three

$\sqrt{3}:3$

All of the root rectangles will have a theme of the same number. So in this case, the root three has a theme of three. You'll later see that the root four has a theme of four, and the root five has a theme of five.

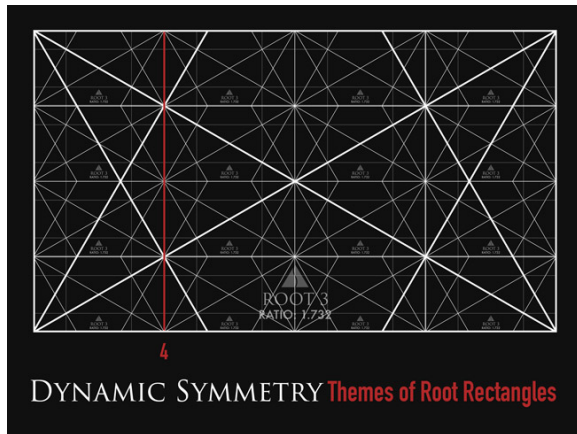


Root Three Rectangle: Theme of Four

$\sqrt{3}:4$

All of the root rectangles will have *at least* one theme higher than their number. I referred to this as the **plus one** earlier. This means the root two can have a theme of three, the root three can have a theme of four, the root four can have a theme of five, and the root five can have a theme of....yep, you guessed it. Six!

We'll even start to notice how the plus one theme is right through the polar points. We'll see more interesting patterns as we analyze further.

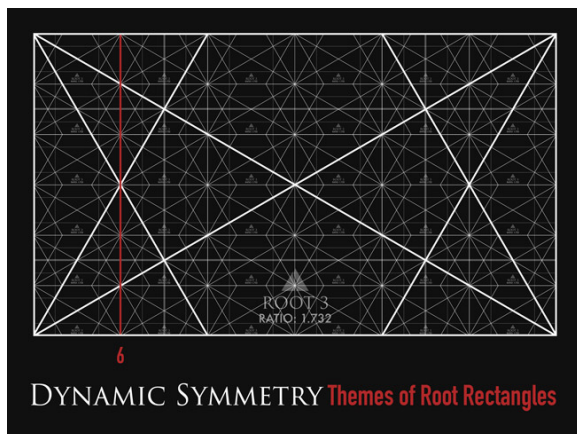


Root Three Rectangle: Theme of Six

$\sqrt{3}:6$

Here we have the root three on the theme of six. See how it's through the middle of the two reciprocals? It's interesting to find that if we multiply the root number (of any root rectangle) by two we get the number for the theme generated by the reciprocal intersection.

For example, we have a root three...multiply three by two, and we will have a theme of six with the location being the intersection of the reciprocals. Take a look at the other root rectangles and you'll see that they break down in this exact way.



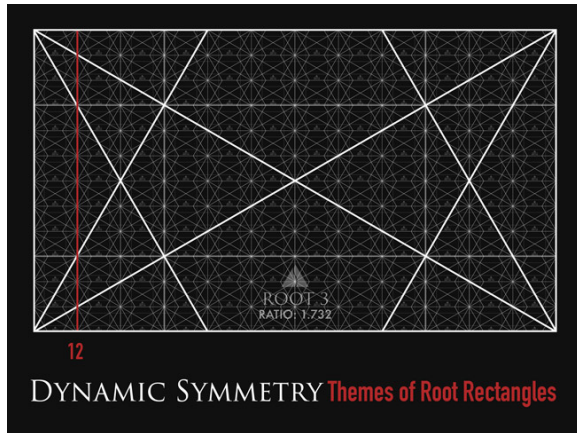
Root Three Rectangle: Theme of Twelve

$\sqrt{3}:12$

Taking what we learned so far, we can find another interesting fact about the numeric dynamic themes. If we take the plus one theme (the one running through the polar points) and multiply it by our root number, we get our last theme. Let's test this out!

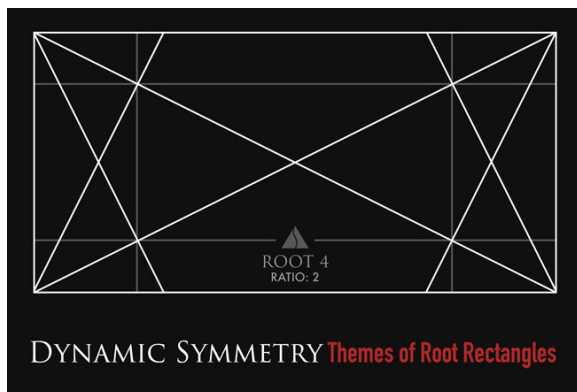
We have a root three rectangle and the plus one is the theme of four. When we multiply four times our root rectangle, three, we get the theme of twelve as seen below. Wow, dynamic symmetry is cool!

When the themes break down this far they almost become unusable unless you are a graphic designer, illustrator, on the computer, or working with a large canvas. Drawing out this many lines would take longer than the painting itself, but as mentioned before, it can be broken down in sections as needed.



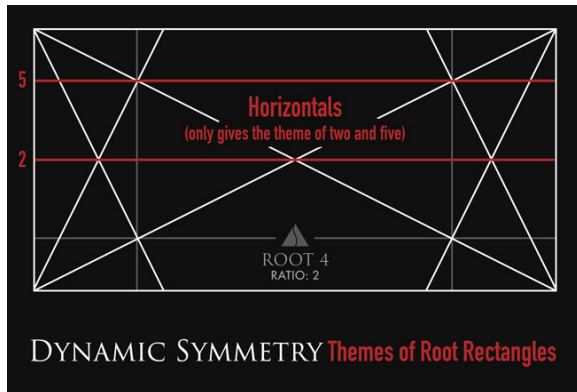
Root Four Rectangle - Ratio: 2

Everything from here on out should be a piece of cake. An interesting thing about the root four rectangle is that it consists of two squares side by side. This is why its ratio is two...every root rectangle starts with the square (see [Day 14](#)).



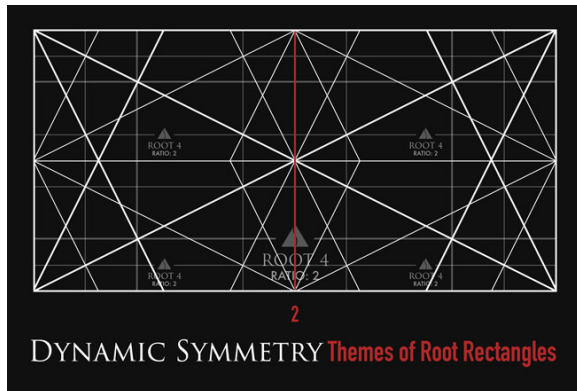
Root Four Horizontal Themes

This time the horizontals give us the themes of two and five. The theme of two, and then the plus one...that's all we get with the horizontals for the root rectangles.

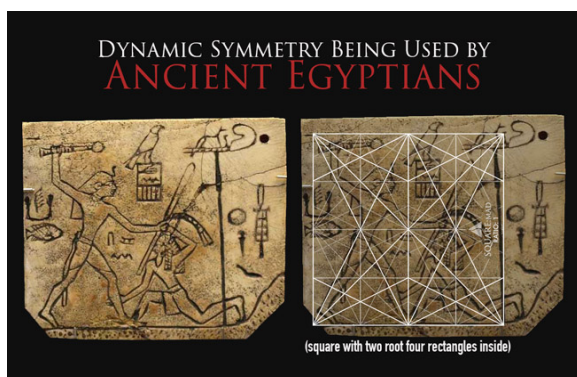


Root Four Rectangle: Theme of Two

$\sqrt{4}:2$



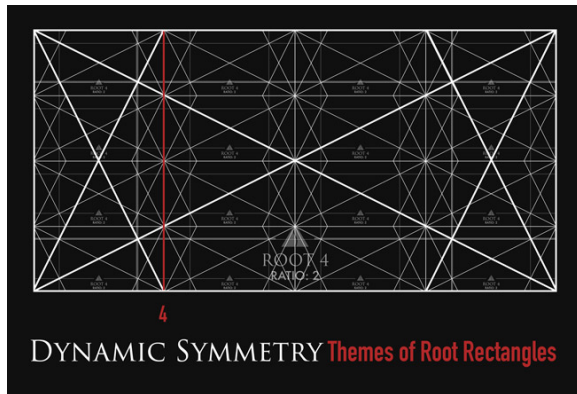
Here's a great example of the Ancient Egyptians using the square with two root 4 rectangles inside. They lock in the characters almost exactly in every way. So, we can clearly see how dynamic symmetry inspired Egyptian art.



Root Four Rectangle: Theme of Four

$\sqrt{4}:4$

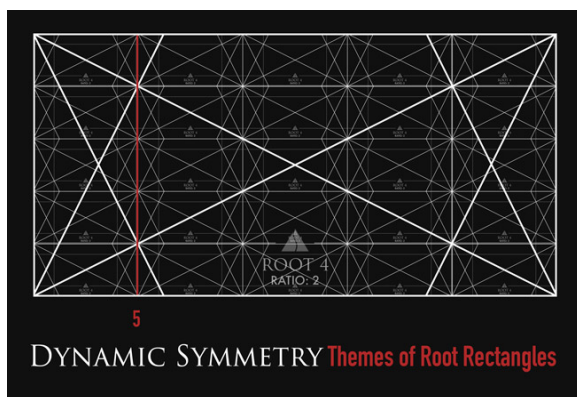
Without looking, can you guess which theme will come next? If we are on a theme of four on a root four, we will get the plus one next.



Root Four Rectangle: Theme of Five

$\sqrt{4}:5$

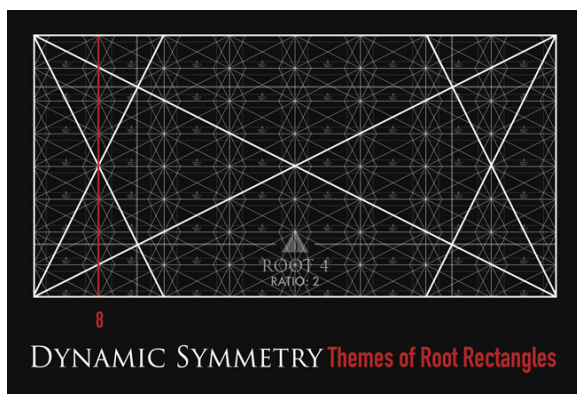
Without looking, can you guess what the last theme will be? If we have a root 4 and we are on the plus one theme...(the one going through the polar points), then we can multiply both numbers together and get the last theme. Four multiplied by five equals twenty. Are you starting to get it by now?



Root Four Rectangle: Theme of Eight

$\sqrt{4}:8$

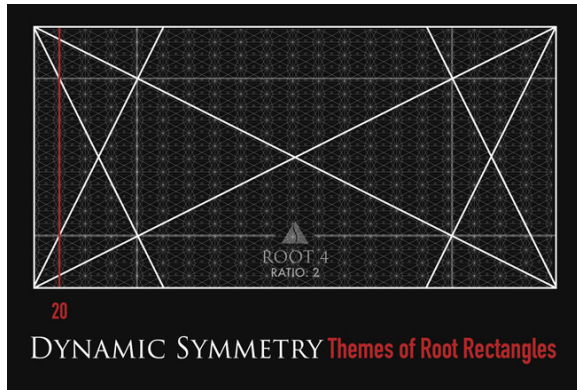
This one should be easy for you. The root number multiplied by two will always give us the theme number of the reciprocal intersection.



Root Four Rectangle: Theme of Twenty

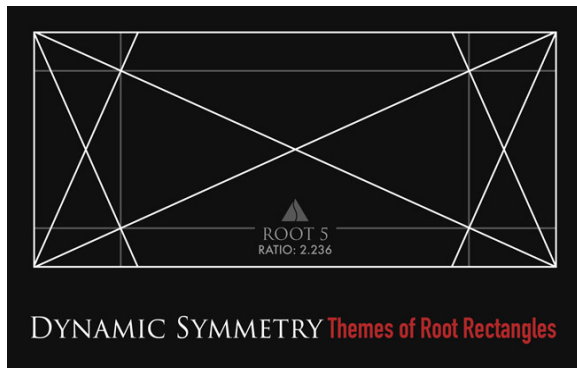
$\sqrt{4}:20$

See, our math added up...plus one multiplied by the root number equals the theme of twenty.



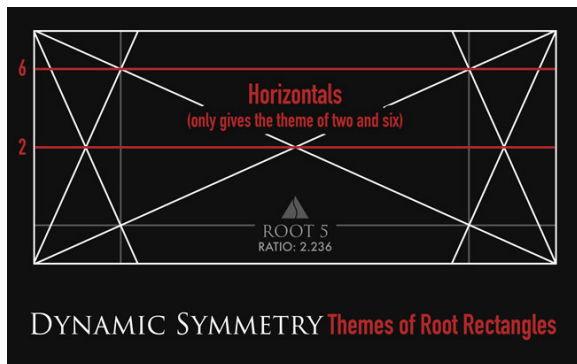
Root Five Rectangle - Ratio: 2.236

Here we have the root five. Interesting fact: it consists of two overlapping phi rectangles which share a square in the middle. Of course, you should already know that if you read [Day 14](#).



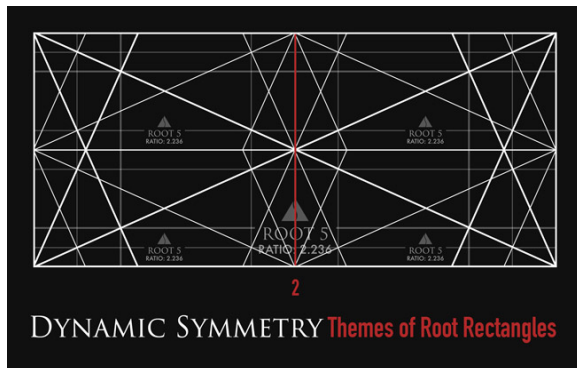
Root Five Horizontal Themes

Remember, we'll get the theme of two and the plus one when we run the horizontals.



Root Five Rectangle: Theme of Two

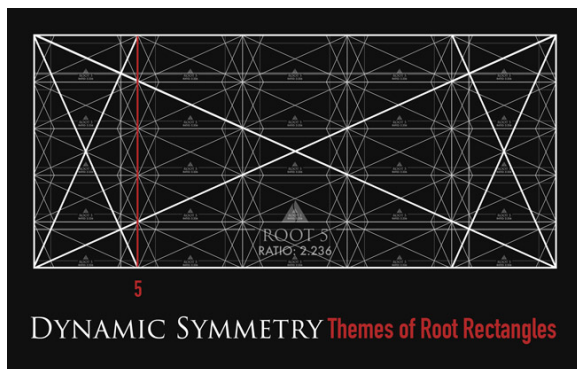
$\sqrt{5}:2$



Root Five Rectangle: Theme of Five

$\sqrt{5}:5$

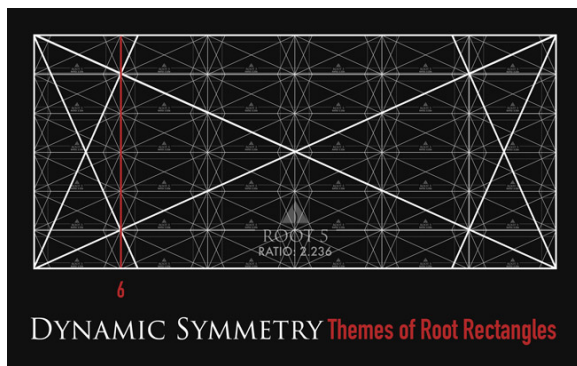
More review: the line dropped from the reciprocal eye at the edge will always have a theme equal to the root number.



Root Five Rectangle: Theme of Six

$\sqrt{5}:6$

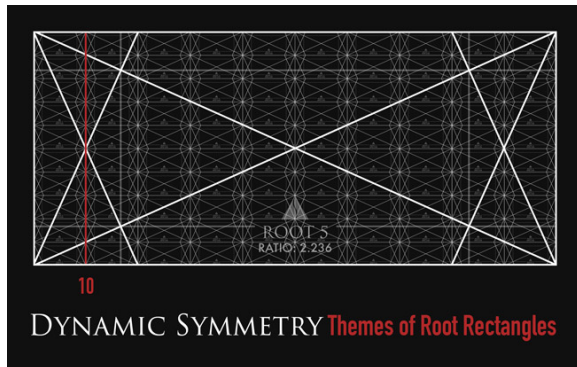
The plus one runs through the polar points. This is getting too easy!



Root Five Rectangle: Theme of Ten

$\sqrt{5}:10$

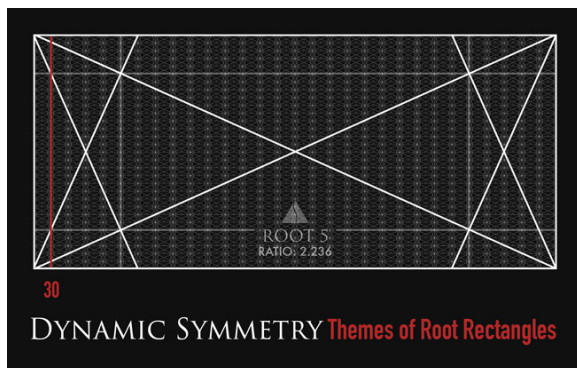
The root number multiplied by two equals the reciprocal intersection.



Root Five Rectangle: Theme of Thirty

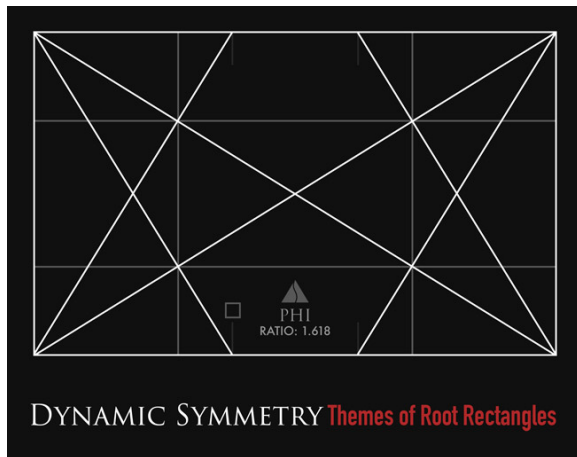
$\sqrt{5}:30$

Take the plus one and multiply it by the root number and it equals our last theme. What a cake walk!



Phi Rectangle - Ratio: 1.618

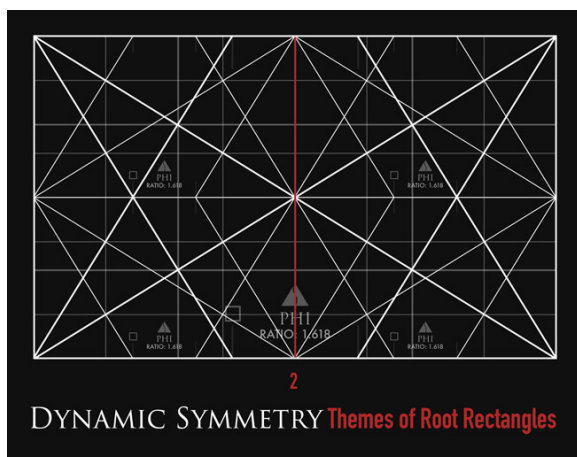
Now, we'll take a look at the rectangles that aren't root rectangles. Here is the ever-popular phi rectangle.



Phi Rectangle: Theme of Two

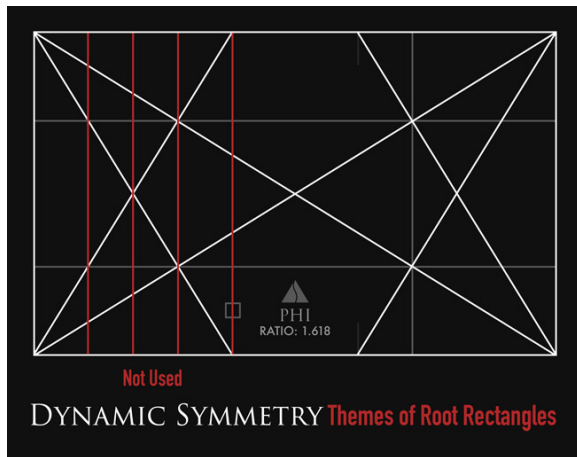
Phi:2

We get the theme of two as mentioned. This is also known as the Major Area Divisions (MAD) that you are already familiar with.



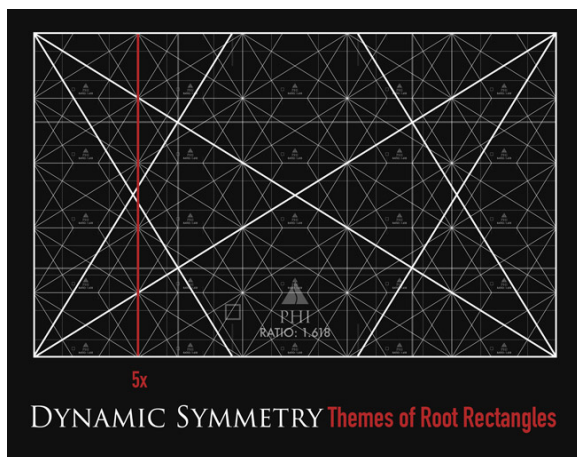
Phi Rectangle Themes Not Used

With the rectangles that aren't root rectangles, we'll have to generate themes that aren't numeric dynamic themes. We can still squeeze five, twelve, or thirty smaller phi rectangles inside the mother, but it will no longer be a NDT. I guess we'll have to give this a name as well...the themes that aren't created by the vertical divisions....

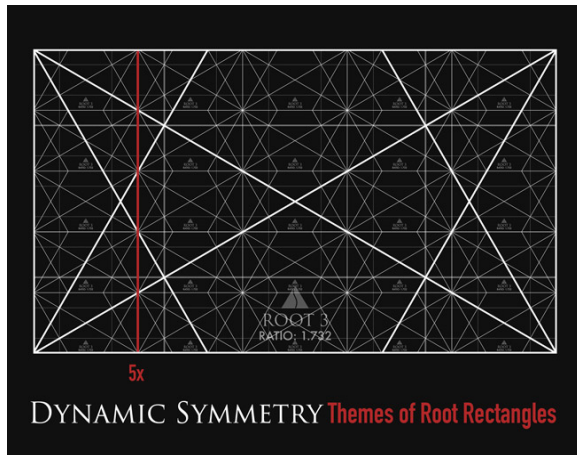


Theme "X"

Here we have a phi rectangle with five smaller phi rectangles inside of it. Since we don't have any numeric dynamic themes past the theme of two, we will have to give this type of theme a name to easily communicate it in a learning environment. We will name this "**theme x**," so it will read as "phi rectangle on the theme of 5x" or written in short "Phi:5x" which will let us know what theme it is on, and that it's not a NDT.

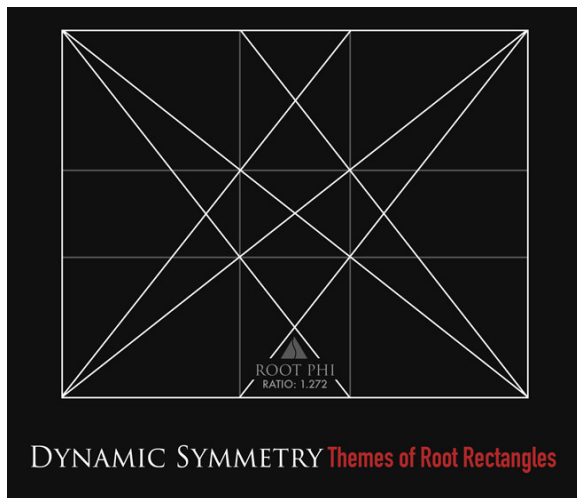


We can also have a root rectangle on the theme of x. Here we have a root three on the theme of 5x ($\sqrt{3}:5x$). Notice how the red line doesn't line up to an intersection point of the basic armature.



Root Phi Rectangle - Ratio: 1.272

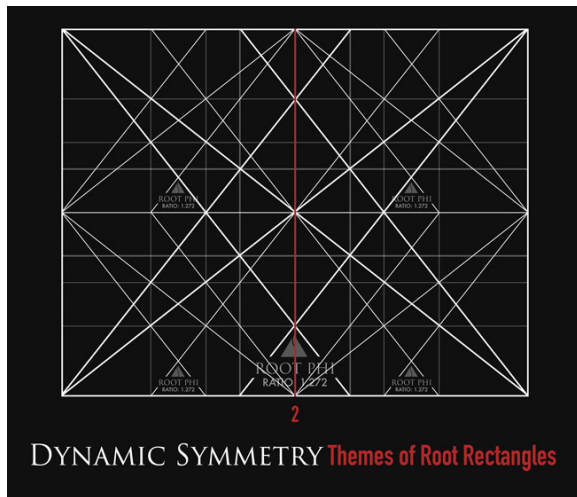
Here we have the root phi...used quite often amongst the master painters. An interesting rectangle where the reciprocals from each side intersect each other.



Root Phi Rectangle: Theme of Two

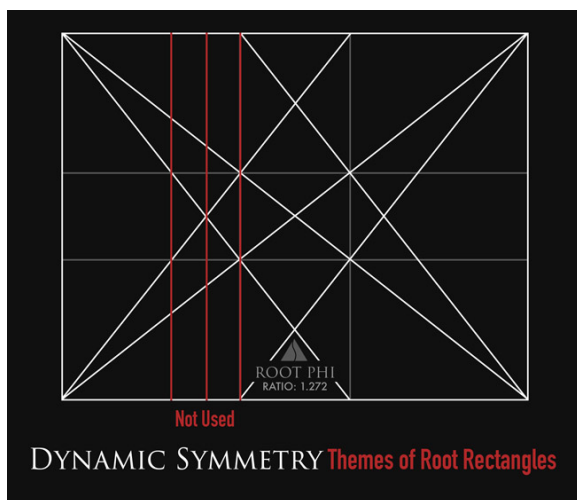
$\sqrt{\phi}:2$

Again, we only have a NDT of two for this rectangle. Even though it has "root" in the name of it, this is a root of phi, not the square.



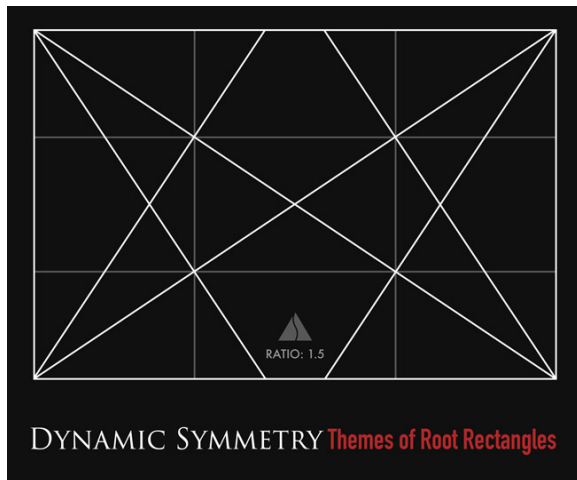
Root Phi Rectangle Themes Not Used

Here we have the vertical intersections, but we can't fit the numeric dynamic themes to them.



1.5 Rectangle - Ratio: 1.5

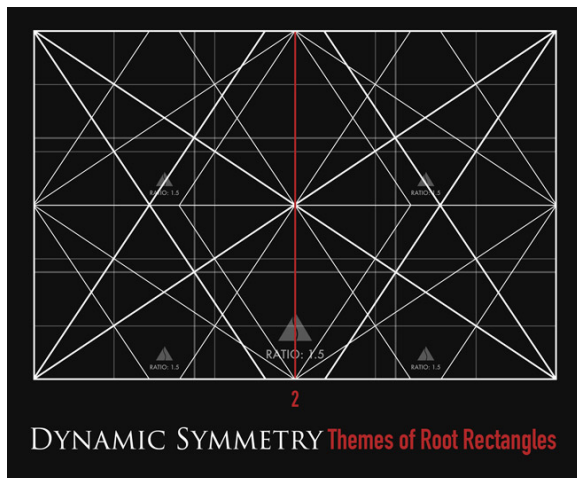
The 1.5 rectangle is a square and a half, and it's also a widely used rectangle. It's equivalent to most camera sensors, and the 24x36 canvas.



1.5 Rectangle: Theme of Two

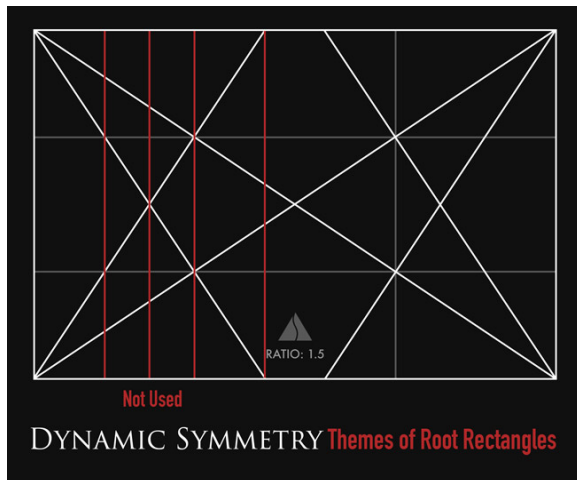
1.5:2

This is a great grid for photographers who are trying to design a set. This can also be called a 1.5 MAD because when we break these rectangles down to the theme of two, we get the **major area divisions**. If we continue to break it down, the area divisions become smaller and it loses some of the **boldness**. Look at the $\sqrt{5}:30$ and you'll see what I mean. It's great to have bold area divisions.



1.5 Rectangle Themes Not Used

We can see that the verticals dropped aren't lining up to a specific theme.



Whew, that was a lot to digest! Come back to this several times if you need...then apply it to your work. Do it on the computer, or draw it out onto your canvas (see [#431](#)).